

The empirical recurrence for OEIS A316951: recovering a conjecture that is not written down

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2 September 2026

Abstract

OEIS A316951 records “Empirical recurrence of order 76”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 87$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 76, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 76 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture one cannot read

OEIS A316951 is “Number of $n \times 6$ 0..1 arrays with every element unequal to 1, 2, 3, 4, 5, 6 or 7 king-move adjacent elements, with upper left element zero.”. It has offset 1 and begins

5, 1493, 71505, 4335826, 259163370, 15440641209, 920789559090, 54906173292280, ...

The entry states:

Empirical recurrence of order 76 (see link above)

As of the “Last modified” line on the live entry (Jul 17 2018 12:23:43, revision 4) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 4096$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 87$ of the 4096, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 87$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 174$ exact terms of the model returns order 76 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{76} c_i a(n-i),$$

c_1	43	c_{20}	20347230820083	c_{39}	−36829368296470546	c_{58}	−26617524824397
c_2	743	c_{21}	−24868441747544	c_{40}	36507957354535392	c_{59}	248398064433
c_3	13121	c_{22}	44923330837877	c_{41}	−28466223789025881	c_{60}	−20825822150269
c_4	92034	c_{23}	−144457436033942	c_{42}	24790711579457228	c_{61}	−18606518625124
c_5	569234	c_{24}	247137940194944	c_{43}	−16082205774729037	c_{62}	−10063804115930
c_6	−199617	c_{25}	−255349518003927	c_{44}	11681022866535878	c_{63}	−2832006211992
c_7	−14940978	c_{26}	823566560164520	c_{45}	−12766594461855143	c_{64}	411360704522
c_8	−87400184	c_{27}	−1708062922065448	c_{46}	10143035565254633	c_{65}	−531997154000
c_9	−114234493	c_{28}	1858910490183129	c_{47}	−3481706582407369	c_{66}	−856261258736
c_{10}	1217365516	c_{29}	−4110367778838452	c_{48}	−4914393846474437	c_{67}	−367097826244
c_{11}	4685384481	c_{30}	8029058292200304	c_{49}	5748761525383854	c_{68}	−71766581848
c_{12}	429313054	c_{31}	−8363998085224345	c_{50}	−1981129614921149	c_{69}	−16473090296
c_{13}	−76339789647	c_{32}	12407963650598815	c_{51}	−772189562708705	c_{70}	−6093336944
c_{14}	−63762372038	c_{33}	−20374166631548612	c_{52}	1502591427424663	c_{71}	−3494067808
c_{15}	166879771439	c_{34}	20769084924666232	c_{53}	−771087703913246	c_{72}	−520816512
c_{16}	1498790841969	c_{35}	−25510217760936263	c_{54}	−679661680379698	c_{73}	512178816
c_{17}	−1727237663180	c_{36}	36854054450901082	c_{55}	300665573348201	c_{74}	161924864
c_{18}	410737387869	c_{37}	−33657659876883304	c_{56}	−179117910760266	c_{75}	7626240
c_{19}	−9266117221072	c_{38}	28495578600475439	c_{57}	−281849864492660	c_{76}	−921600

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 76, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $76 + 4$ terms removed returns order 76 as well, so $\deg q_{\min} = 76 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 14 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $76 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 76 that this sequence satisfies — and that family turns out to have exactly one member.

References

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